Round 7 — The whole error is three numbers per neuron. How do we compute them better, cheaply?

Bias-free ReLU MLP n=1024, L=16. Score = MSE × max(0.1, FLOPs/2^{41}). 1 unit = one dense n^3 product = 2^{31} FLOPs. V29 baseline: 260 u → raw MSE 2.1e-8. Claims: D=derived, C=conjecture, M=measure-this.

Q1 — Variance channel (worth −47 % of final MSE)

Pre-activation variance of neuron i at the next layer is the quadratic form $v_i = w_i^T C_{post} w_i$. Only the 300–400 neurons with $|\mu/\sigma| < 0.75$ matter, and they need ~0.1 % relative accuracy. 95 % of the chain's variance error is a random projection of the previous layer's off-diagonal post-ReLU covariance error; that error has a fresh component of ~4.2e-5 rms per entry that the chain's own closure explains only 3–8 % of.

Proposal 1a — Adjoint / selected-row recursion for the needed quadratic forms (D)

State kept: the full post-ReLU covariance C_post (already present) plus a thin matrix $R \in \mathbb{R}^{n\times m}$ whose columns are the weight rows w_i of the m ≈ 350 "active" neurons (those with $|\mu/\sigma| < 0.75$ at the next layer). Size: $n^2 + n\,m \approx n^2 + 0.34\,n^2$.

Update / evaluation: the needed variances are simply the m diagonal entries of $R^T C_{post} R$. Forming $R^T C_{post}$ costs one (m × n) @ (n × n) matmul = $m\,n^2$ (2n−1) FLOPs ≈ 0.34 units; the subsequent (m × n) @ (n × m) is only 0.12 units. Total ≈ 0.46 units per layer instead of a full covariance transport through the ReLU closure (≈ 2–3 units). Because C_post is already computed by the chain, the only extra work is the thin matmul.

This does not reduce the error in C_post itself; it only avoids paying for the quadratic forms of the inactive neurons. The error reduction must come from a better closure for the off-diagonal of C_post (see 1b).

Falsifier (stored arrays, <5 min): extract the true active set from the teacher ($|\mu/\sigma| < 0.75$); form R from the true next-layer weight rows; compute $\mathrm{diag}(R^T C_{chain} R)$ and compare to the true pre-activation variances on those neurons. Relative error must match the chain's existing 0.15 % (sanity check). Then replace C_chain by C_true and verify that the quadratic forms become exact to machine precision.

Proposal 1b — Missing off-diagonal covariance closure term (C)

The fresh ~4.2e-5 rms per entry is two orders of magnitude larger than the Monte-Carlo noise of the 6e7-sample teacher. The chain's post-ReLU C is accurate on the diagonal and on K22, so the missing piece is a third-order (or higher) contribution to the off-diagonal that the bivariate Wick/Edgeworth program drops.

Leading candidate (Edgeworth expansion of the bivariate density up to order 4, retaining the mixed cumulant $\kappa_3(i,i,j)$ that the chain already carries):

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δC_post,ij ≈ (1/2) Φ_i' Φ_j' · κ₃(i,i,j) · (μ_i / σ_i²) + cyclic(i↔j)
+ (1/6) Φ_i'' Φ_j · κ₄(i,i,i,j) + \ldots
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All ingredients (Φ, Φ', Φ'', κ_3 slice, $\kappa_4(i,i,i,j)$ if available) are already in the chain or can be obtained from the stored teacher tensors. Cost of the correction: one Hadamard product of the κ_3 slice with a rank-1 matrix of size n×n → ≈ 0.01 units.

Falsifier (<15 min): at layers 6,10,14 form the residual Δ = C_true_post − C_chain_post (off-diagonal). Regress Δ against the candidate term above (and the cyclic and κ_4 terms) on the stored arrays. If $R^2 > 0.4$ and the coefficient is stable across layers, the term is real; inject it into the chain and measure the drop in final-layer variance error (target: from 0.15 % to < 0.05 %).

Proposal 1c — Control-variate Monte-Carlo for the variance correction only (C)

Let v_chain,i be the chain's estimate of the pre-activation variance. The error $\delta v_i = v_{true,i} - v_{chain,i}$ has empirical rms ~0.15 % of v. A stratified / antithetic Monte-Carlo that samples only the residual can be variance-reduced by using the chain's own Gaussian surrogate as control variate:

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δv^_i = (1/S) Σ_s [ (w_i^T h_post^{(s)})² − (w_i^T h_G^{(s)})² ] + v_G,i − v_chain,i
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where h_G is drawn from the chain's Gaussian N(μ,C). Because the leading quadratic is identical, the difference has variance proportional to the non-Gaussian part only. Empirical estimate from the teacher: the residual variance is ~1/40 of the naïve variance, so S ≈ $2.5\cdot10^4$ samples give 0.1 % accuracy on the active neurons. Cost: $2.5\cdot10^4$ × 16 layer-forwards ≈ 400 units if done naïvely, but a single forward through all layers already produces all post-ReLU activations; with antithetic pairs and reuse of the same samples for every active neuron the cost falls to ~8–12 units for the whole network (one-time, not per layer).

Falsifier: on the stored teacher samples compute the residual variance of the control-variate estimator versus naïve MC for S = 10^3\ldots$10^5$; confirm the ~40× reduction. Then inject the estimated δv into the mean closure and measure final MSE.

Q2 — κ_4 diagonal channel (worth −30 %)

$\kappa_4(z_i')$ of the next layer receives contributions from every post-ReLU cumulant tensor of order ≤4. The (4) and (2,2) contractions set the overall level (slope 1.00 when the true slices are used); the per-neuron pattern that the chain misses lives in the (2,1,1) and (1,1,1,1) parts. Those parts are regenerated at every ReLU from the pre-activation $\kappa_3 \otimes C$ and κ_4 content; linear transport of old births accounts for only ~10 % of the deep diagonal.

Leading (2,1,1) / (1,1,1,1) content at a ReLU (D)

Write the pre-activation vector z ~ (approximately) a Gaussian with a third- and fourth-order Edgeworth correction. The post-ReLU fourth cumulant of a single coordinate h = relu(z) has the exact moment expressions (obtainable by differentiating the moment-generating function or by Stein's lemma under the Gaussian measure):

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κ₄(h_i,h_i,h_j,h_k) |_{j≠k} ≈ Φ_i''' · C_ij C_ik + Φ_i'' · (κ₃(i,j,?) terms) + \ldots
κ₄(h_i,h_j,h_k,h_l) (all distinct) ≈ Φ_i' Φ_j' Φ_k' Φ_l' · (products of C's) + higher.
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After collecting the dominant terms that survive contraction against four copies of a weight row w, the contribution to the next-layer κ_4 diagonal that carries the missing pattern is

$$\delta\kappa_4(z'_r) \approx 6\,\Sigma_{i\neq j}\, w_{ri}^2\, w_{rj}^2 \cdot \kappa_4(h_i,h_i,h_j,h_j)$$
$$+\ 4\,\Sigma_{i\neq j}\, w_{ri}^3\, w_{rj} \cdot \kappa_4(h_i,h_i,h_i,h_j)$$
$$+\ 24\,\Sigma_{i<j<k}\, w_{ri}\, w_{rj}\, w_{rk} \cdot (\text{3-point post-ReLU cumulants}) \cdot \ldots$$

The first sum is a quadratic form in the matrix $K22_post = (\kappa_4(h_i,h_i,h_j,h_j))_{ij}$; the second is a cubic form in the tensor slice $\kappa_4(i,i,i,j)$. Both slices are already stored by the teacher and can be maintained approximately by the chain.

Cheap evaluation without the full tensor (C)

State: the two matrices / slices already present (or cheaply regenerated) $K22_post \in \mathbb{R}^{n\times n}$ and $K31_post \in \mathbb{R}^{n\times n}$ (the (3,1) slice). Size $2n^2$.

Evaluation for every neuron r:

$$\delta\kappa_4_r = 6\,(w_r^2)^T\, K22_post\, (w_r^2) + 4\,(w_r^3)^T\, K31_post\, w_r$$

where $w_r^2 = w_r \circ w_r$, $w_r^3 = w_r \circ w_r \circ w_r$ (element-wise). For a single r the cost is two matrix-vector products with the dense K22 / K31 plus O(n) Hadamards $\approx 4\,n^2$ FLOPs. For all n neurons a naïve loop is $4\,n^3 = 4$ units. A better organisation: form the matrix W2 whose rows are w_r^2, then $\mathrm{diag}(W2\ K22\ W2^T)$ can be computed by a single $(n\times n)@(n\times n)$ product followed by a Hadamard and row-sum, total ≈ 2.1 units for the (2,2) piece; the (3,1) piece is analogous and costs another ~ 2 units. Grand total ≈ 4.2 units per layer if done for every neuron; restricted to the ~ 350 active neurons it drops to ≈ 1.5 units.

Falsifier (<20 min): at layers 8–14 form the true $\delta\kappa_4$ from the teacher's full post-ReLU κ_4 tensor contracted with the true next-layer weights. Compare with the expression above that uses only K22 and K31. If the correlation with the true per-neuron pattern exceeds 0.7 and the residual is uncorrelated with the chain's current κ_4 error, inject the term into the mean closure and measure the final MSE drop.

Q3 — Sampling the three corrections

Naïve Monte-Carlo needs $\sim 10^6$ samples for the variance correction and $\sim 10^5$ for κ_4 — far above the 10^4–10^5 budget that costs only a few units. The question is whether a control variate built from the chain's own Gaussian (or Edgeworth) surrogate can cut the variance by $\sim 100\times$.

Control-variate estimator (D/C)

Let $\theta \in \{\mathrm{var}, \kappa_{3iii}, \kappa_{4iiii}\}$. Write

$$\hat{\theta}_i = \theta_{chain,i} + (1/S)\,\Sigma_s\,[\ \theta(h^{(s)})_i - \theta(h_G^{(s)})_i\ ]$$

where $h^{(s)}$ is a true forward sample and $h_G^{(s)}$ is drawn from the chain's Gaussian $N(\mu,C)$ (or from the chain's Edgeworth-corrected marginals). The difference inside the sum has mean equal to the true correction and variance equal to the residual non-Gaussianity that the chain missed.

From the teacher (6e7 samples) the residual variance for the variance channel is $\sim 1/40$ of the naïve variance (already used in 1c). For the κ_4 diagonal the residual variance is larger — roughly 1/8 of naïve — because the chain's κ_4 regeneration is only moderately correlated (0.75–0.83). For κ_{3iii} the residual is $\sim 1/15$ of naïve (the slice is already excellent). Consequently, to reach the accuracy needed for a 2× reduction of the corresponding error term one needs approximately

$$S_var \approx 2\cdot 10^4,\ S_\kappa_4 \approx 1.2\cdot 10^5,\ S_\kappa_3 \approx 4\cdot 10^4$$

forward samples. At 0.016 u per full-network sample the budgets are ~ 0.3 u, ~ 2 u and ~ 0.6 u respectively — all well inside the "few units" regime. Antithetic pairs (x and −x, possible because the network is 1-homogeneous up to the absolute-value of the first-layer norms) give another ~ 1.5–$2\times$ reduction, bringing κ_4 within ~ 1 u.

Rao–Blackwellisation over the last weight layer: given the post-ReLU activations of layer L−1, the final-layer pre-activation cumulants are exact polynomial functions of those activations and the last weight matrix. One can therefore average the last-layer contribution analytically (or with a second, cheap MC over the last W only). This removes the variance of the last layer entirely and is free once the L−1 activations are stored. Expected additional variance reduction: $\sim 1/L \approx 6$ % (modest but free).

Falsifier (<30 min): on the stored teacher samples compute the empirical variance of the control-variate estimators for each of the three channels at $S = 10^3, 10^4, 10^5$. Confirm the reduction factors above. Then run a true sampling experiment with $S = 3\cdot 10^4$ antithetic pairs, inject the three estimated corrections into the mean closure, and measure final MSE (target: from 2.1e-8 toward 0.5–0.8e-8).

Q4 — Ranking and first experiment

Expected score gain per unit of compute (rough, from the oracle table and the costs derived above):

Channel	Oracle gain	Cheapest credible cost	Gain / unit
Variance (1b+1a)	−47 %	~0.5 u (analytic term) or ~10 u (CV-MC)	highest
κ_4 diagonal (Q2)	−30 %	~1.5–4 u analytic	high
κ_3 diagonal	−14 %	already excellent; sampling ~0.6 u	low
All three via CV-MC	−94 % (oracle)	~2–3 u total	very high if variance reduction holds

First experiment to run (highest expected information per CPU-hour): the residual regression of Proposal 1b. At layers 6, 10, 14 form $\Delta = C_true_post - C_chain_post$ (off-diagonal) and regress against the candidate Edgeworth term that uses the already-available $\kappa_3(i,i,j)$ slice and the Gaussian density factors Φ, Φ'. Cost: a few matrix multiplies on stored arrays, <15 min. If $R^2 > 0.4$ the term is real and can be inserted into the chain at almost zero extra FLOPs, immediately capturing a large fraction of the −47 % variance oracle. If R^2 is low, the same residual immediately tells us which higher-order mixed cumulant

is missing, giving a concrete target for the next analytic term.

Second priority: the analytic (2,1,1)/(1,1,1,1) contraction of Q2 on the active neurons (~1.5 u). Third: a small control-variate pilot ($S = 10^4$) that measures the realised variance reduction for all three channels simultaneously; if the ~40× / 8× / 15× factors hold, a 2–3 unit sampling post-processing step can approach the oracle floor.

Summary: the entire error of V29 lives in three per-neuron numbers. The variance channel is both the largest (47 %) and the cheapest to attack (a single missing off-diagonal Edgeworth term that uses quantities already present). The κ_4 pattern is the next largest and admits an analytic contraction at ~1.5 u on the active set. Sampling corrections are viable at 2–3 u total if the measured residual variances hold. Nothing in the closed list of previous rounds is re-proposed.